

Effects of Progestins on Bone Metabolism in Postmenopausal Women

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Medroxyprogesterone acetate (MPA) is the progestin used most commonly in postmenopausal women to attempt to reduce the risk of endometrial cancer associated with estrogen replacement and, more recently, as the sole agent for the treatment of hot flashes. The effects of this progestin were examined on the fasting urinary ratios of calcium (Ca) and hydroxyproline (OHPr) to creatinine (Cr) as indices of bone resorption. MPA, 20 mg, significantly reduced Ca/Cr and OHPr/Cr to levels similar to those in premenopausal women. The effects of MPA were eightfold greater than those of megestrol acetate (MA) as previously reported. The effects of ethinyl estradiol were greater than those of both MPA and MA. Long-term studies

of bone density will be required to confirm these apparent, beneficial effects of MPA on bone metabolism in postmenopausal women.

Prevention of bone loss and subsequent osteoporosis is the single most important indication for postmenopausal hormonal replacement.⁹ Progestins are being utilized increasingly in postmenopausal women to reduce the risk of endometrial cancer during estrogen-replacement therapy.^{5,7} In women with a prior history of estrogen-dependent neoplasia or other contraindications to estrogen therapy, progestins may be used for extended periods of time for the control of menopausal hot flashes.^{1,3,12} We examined the effects of the most commonly utilized progestin, medroxyprogesterone acetate (MPA), on the bone metabolism of postmenopausal women. In order to assess the dose required to affect bone metabolism, a dose-response study was carried out. The dosage providing the maximal effect was then examined in a double-blind crossover study.

Materials and Methods

Dose-Response Studies

We studied 15 postmenopausal women before treatment and following the daily oral administration of 10 and 20 mg of MPA and 20 µg of ethinyl estradiol (EE). The results were compared with the values in 20 untreated premenopausal women and 10 postmenopausal women treated with megestrol acetate (MA), 20, 40 and 80 mg, that had been reported previously.^{3,11} Each dose of medication was given for four weeks.

Double-Blind Crossover Study

On the basis of the results of the above, 20 more postmenopausal women were studied. After baseline measurements the subjects were randomized into two equal groups. One group received 20 mg daily of MPA, and the other group received placebo tablets. At the end of four weeks of treatment each group received the alternate medication for four more weeks. Neither the subjects nor the investigators were aware of which tablets were active and which placebo.

Assessment of Bone Metabolism

Before treatment and after each four-week course of medication or placebo the subjects were instructed to take nothing by mouth after 8 P.M. At 8 A.M. the following day the subjects voided and then drank 250

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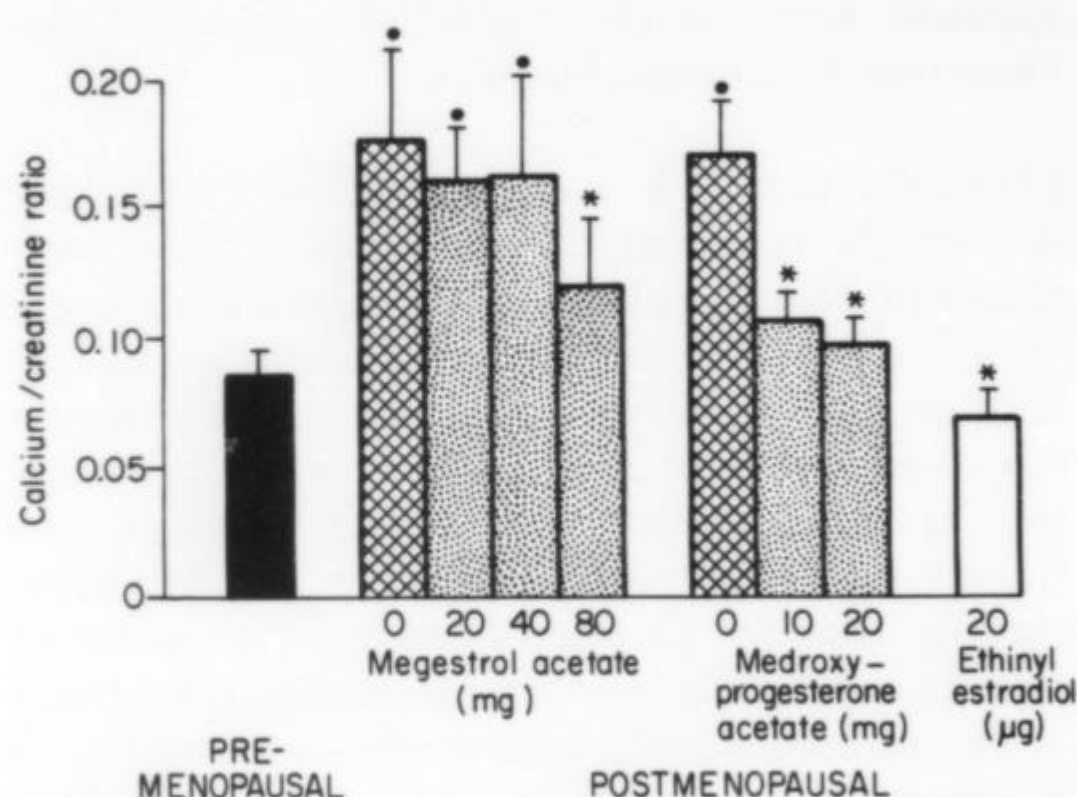


Figure 1

Mean $\pm$ SE levels of the Ca/Cr ratio in the premenopausal women and in the postmenopausal subjects before treatment and at the end of four weeks of megestrol acetate, 20, 40 and 80 mg; medroxyprogesterone acetate, 10 and 20 mg; and ethinyl estradiol, 20 µg. Each dot (•) above a bar indicates a significant difference from the values in premenopausal women. An asterisk (*) denotes a significant difference from the pretreatment value.

ml of distilled water. After 60 minutes a urine sample was collected and stored at -20°C until analysis.

Urine specimens were analyzed for the concentrations of calcium (Ca) by atomic absorption, hydroxyproline (OHP) by a modification of the method described by Blumenkrantz^{2,3} and creatinine (Cr) by autoanalyzer (Technicon Instrument Co., San Francisco, California). The urinary Ca/Cr ratio was calculated by dividing the Ca concentration in milligrams per 100 milliliters by the Cr concentration in milligrams per 100 milliliters. According to the work of Nordin and co-workers, in the fasting state urine Ca mainly reflects Ca coming from bones.¹⁴ Based on this observation, the Ca/Cr ratio was used as an index of bone resorption. When collagen is destroyed, most of it is hydrolyzed into its component amino acids. Hydroxyproline is one of them, and its excretion in the urine reflects the resorption of bone matrix.⁴ Measurement of OHP/Cr was also used as an index of bone resorption.

Statistical analysis was performed using appropriate paired and unpaired Student's t-tests. The cross-over study was analyzed by a repeated measures analysis of variance using a calculation for each data point as a change from the baseline.

Results

Figure 1 shows the mean $\pm$ SE levels of Ca/Cr in the

premenopausal subjects and in the postmenopausal women before treatment and after each dose of MA, MPA and EE. In the postmenopausal subjects given MA the pretreatment ratio of Ca/Cr (0.17 ± 0.03) was significantly ($p < 0.01$) higher than the ratio observed in the premenopausal women (0.09 ± 0.01). A significant ($p < 0.05$) reduction of Ca/Cr was noted only at a dose of 80 mg of MA. In the postmenopausal women who received MPA the pretreatment ratio of Ca/Cr (0.159 ± 0.02) was significantly ($p < 0.02$) higher than the mean ratio in the premenopausal women. Significant reductions ($p < 0.01$) of Ca/Cr were noted during the administration of both 10 and 20 mg of MPA. The mean levels of Ca/Cr during 80 mg of MA and 10 and 20 mg of MPA were not significantly different from the mean Ca/Cr in the premenopausal women. During EE treatment the ratio of Ca/Cr (0.069 ± 0.01) was significantly ($p < 0.001$) reduced, to a level that was significantly ($p < 0.01$) lower than that observed in the postmenopausal subjects who received the highest doses of MA and MPA.

Figure 2 shows the corresponding levels of OHP/Cr. The level of OHP/Cr was significantly ($p < 0.01$) elevated in the postmenopausal subjects who received MA. No significant changes of OHP/Cr were observed at any of the doses studied. The ratio of OHP/Cr was not significantly increased in the postmenopausal subjects given MPA and did not change

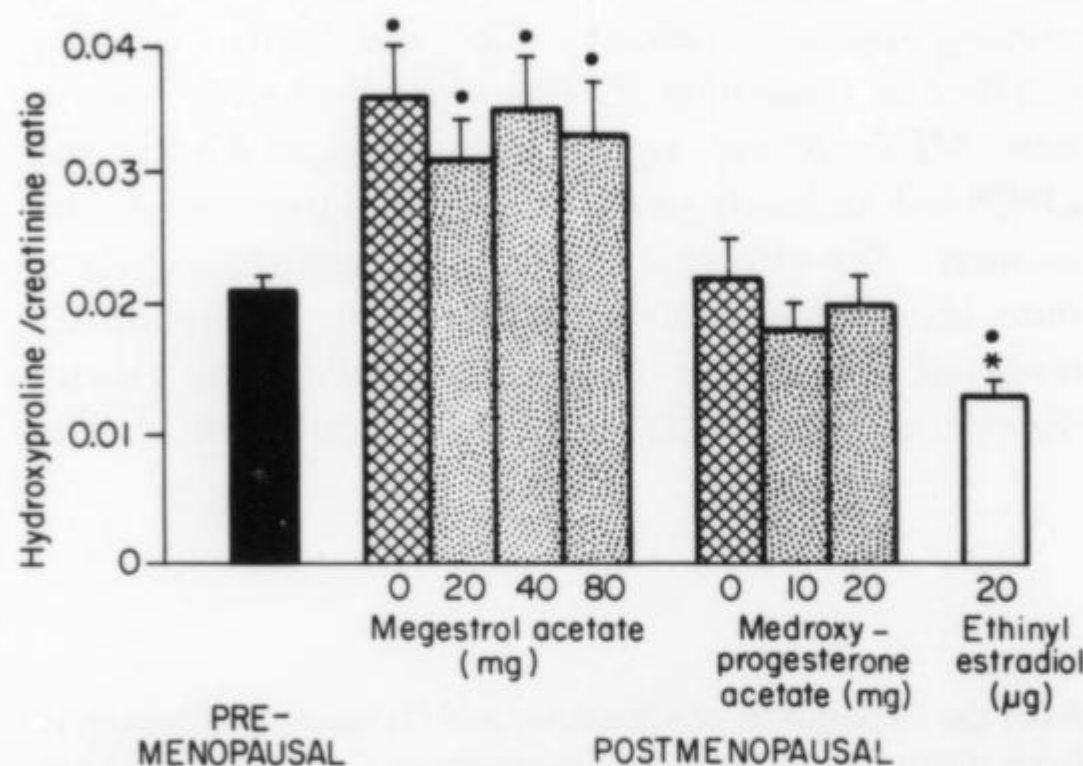


Figure 2

Mean $\pm$ SE levels of the OHP/Cr ratio in the premenopausal women and in the postmenopausal subjects before treatment and at the end of four weeks of megestrol acetate, 20, 40 and 80 mg; medroxyprogesterone acetate, 10 and 20 mg; and ethinyl estradiol, 20 µg. Each dot (•) above a bar indicates a significant difference from the values in premenopausal women. An asterisk (*) denotes a significant difference from the pretreatment value.

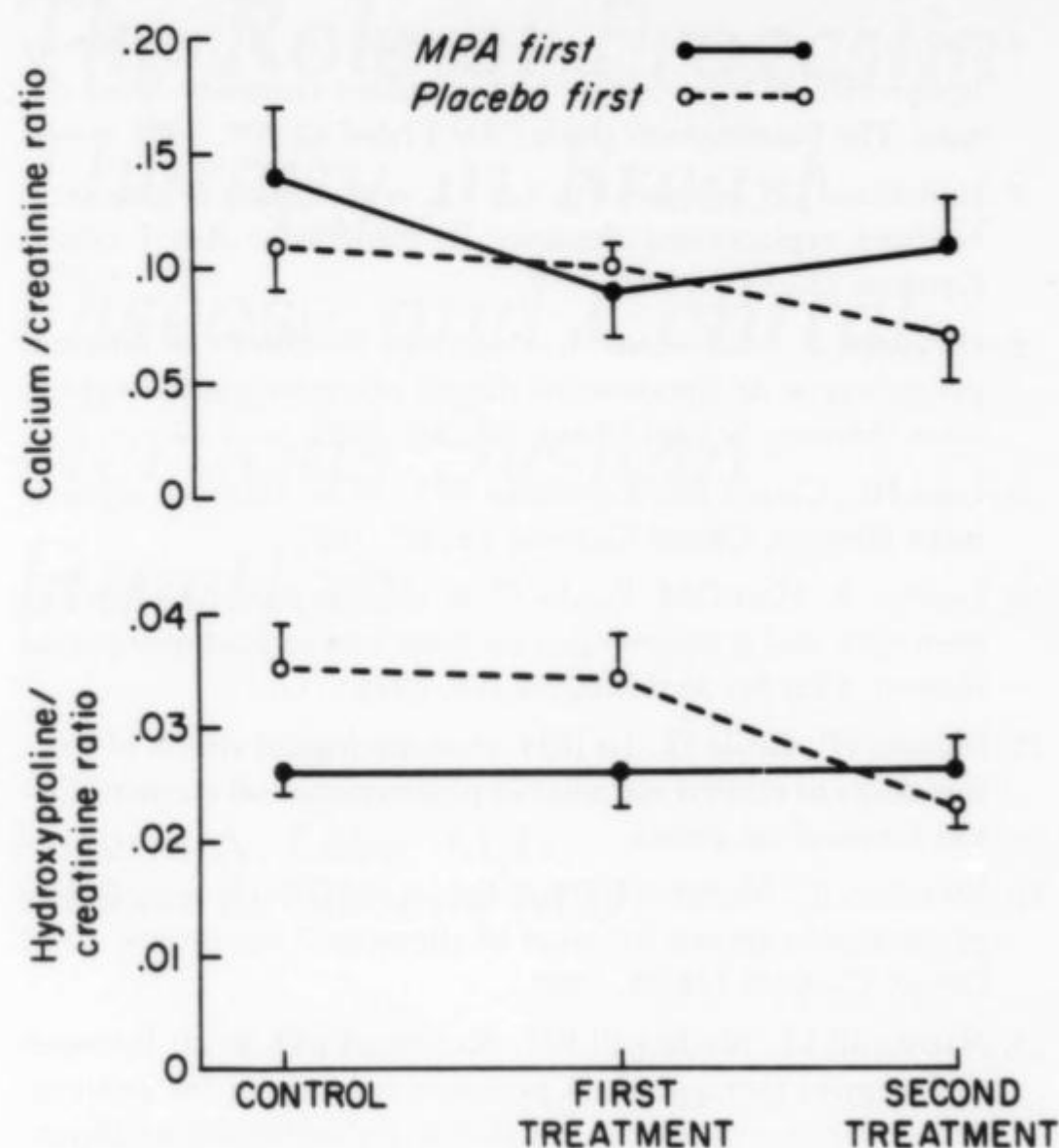


Figure 3
Mean $\pm$ SE levels of the Ca/Cr ratio and the OHPr/Cr ratio before treatment and during administration of either MPA or placebo in a double-blind, crossover study.

with treatment. A significant decrease ($p < 0.01$) in OHPr/Cr was noted during EE treatment, to a level that was significantly ($p < 0.001$) lower than in the premenopausal women.

Figure 3 shows the ratios of Ca/Cr and OHPr/Cr in the double-blind crossover study of 20 mg of MPA as compared with the placebo. In the women who received MPA during the first treatment period Ca/Cr was reduced, although this change was not statistically significant. Subsequent administration of placebo resulted in a rise of Ca/Cr toward the baseline. In the women who received the placebo first, no change of Ca/Cr was noted. During the second treatment period MPA administration was associated with a significant ($p < 0.05$) reduction of Ca/Cr. When both groups were analyzed together a significant decrease ($p < 0.05$) of Ca/Cr was noted with MPA but not with placebo administration. In the subjects who received MPA first, OHPr/Cr prior to treatment was not significantly greater than in the premenopausal women and was unchanged during MPA and placebo administration. In the women who received placebo first, OHPr/Cr was significantly elevated prior to treatment, unchanged during placebo administration and

significantly ($p < 0.02$) reduced during MPA treatment. When both groups were analyzed as a whole a significant ($p < 0.01$) decrease in OHPr/Cr was noted with MPA treatment but not with placebo administration.

Discussion

The present study showed a significant reduction in the fasting excretion of both calcium and hydroxyproline during the oral administration of 20 mg of MPA daily. These data suggest that MPA may reduce bone resorption in postmenopausal women. Long-term studies of bone density will be required to determine whether MPA has a protective effect on bone.

A comparison of the effects of MPA with those of MA, as previously published,³ showed that 10 mg of MPA reduced Ca/Cr to a degree similar to that of 80 mg of MA. The reason for this eightfold difference in potency is not apparent since the structures of these two progestins are similar. No reduction of OHPr/Cr was observed with MA.

The effects of EE on Ca and OHPr excretion were greater than those of both MPA and MA, suggesting that estrogen in adequate doses may provide a greater degree of protection to bone than the doses of MPA and MA studied. These data suggest that estrogen therapy may be the preferred method of hormone replacement for the prevention of osteoporosis. For certain women at increased risk with estrogen treatment progestin therapy may provide a more favorable risk:benefit ratio than does estrogen replacement.

Currently it is commonly recommended that one administer a progestin for ten or more days during each cycle of postmenopausal estrogen replacement. The data from the present study suggest that the use of MPA or MA in conjunction with estrogen-replacement therapy is unlikely to result in greater calcium excretion than with estrogen alone, although experimental verification of this suggestion should be carried out. It is possible that an additive or even synergistic effect on the suppression of calcium excretion might occur with the addition of a progestin to estrogen-replacement regimens, thus allowing the use of smaller doses of estrogen.

Other progestins of the 19-nortestosterone group have also been shown to reduce calcium excretion⁴ and retard the loss of bone density¹⁰ in postmenopausal women. However, this group of progestins has been shown to reduce high-density-lipoprotein

cholesterol (HDL-C), whereas MPA has no effects on the level of this lipoprotein fraction.⁸ Women with lower levels of HDL-C have a higher incidence of cardiovascular disease.⁶ The present data, showing the favorable effects of MPA on bone metabolism, and experience with the protective action of this progestin in regard to endometrial⁵ and possibly breast¹³ cancer further support its use as the progestin of choice for postmenopausal hormone replacement.

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